Machine Learning Analysis of a Multimodal Health Dataset for Predicting Loneliness

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CSC 350 Project 4

***Abstract*—This project takes a multimodal health dataset collected from 39 participants over 28 days and applies it to predict loneliness-related outcomes. The dataset includes smartphone behavior, wearable device data, smartwatch sensor data, and survey responses. We extracted participants' behavioral, physiological, and psychological features, and compared four supervised learning models: Support Vector Machine (SVM), Random Forest, XGBoost, and a Neural Network. Random Forest resulted with the strongest held-out performance, demonstrating an F1 score of 0.8889 and an accuracy of 0.8750. Overall, the results suggest that multimodal signals, particularly ecological momentary assessment features and sleep-related physiological measures, carry useful information about loneliness and mental well-being.**

I. Dataset Understanding

The purpose of this dataset is to support research on loneliness and well-being through multimodal behavioral and physiological data. It comes from the study "Loneliness and Well-being in Finnish Immigrants: A Multimodal Dataset from Wearables and Passive Data Collection" and was created to explore whether passive sensing and self-reported measures can help describe mental-health outcomes.

The dataset includes four main types of data. First, smartphone data collected with the AWARE app capture screen use, notifications, calls, messages, and related behavior. Second, Oura Ring data provide sleep duration, sleep quality, activity summaries, heart rate, and heart-rate variability. Third, Samsung Watch files include smartwatch sensor information such as heart rate and movement-related signals. Lastly, the survey and psychological measures include PHQ-9, perceived stress, social connectedness, UCLA loneliness, and ecological momentary assessment responses.

This kind of dataset has potential to support several research applications. It could be used to predict loneliness risk, study links between sleep and emotional well-being, examine how smartphone and wearable behavior relate to social isolation, and compare how much different sensing modalities contribute to mental-health prediction.

II. Data Preprocessing

The project includes all 39 participants. We first organized each raw file by participant and modality, then converted them into participant-level features for modeling. Timestamps stored in milliseconds were converted to dates for uniformity and easier reading, and then we computed modality-specific daily aggregates before condensing them into participant summaries.

Behavioral features from the AWARE data included daily screen-on time, screen event counts, call counts, call duration, missed calls, message counts, and notification counts. Oura features summarized sleep duration, sleep efficiency, sleep score, steps, activity score, readiness score, heart-rate variability balance, RMSSD, and average heart rate. Samsung Watch data were aggregated into mean heart rate, a heart-rate variability proxy based on daily heart-rate standard deviation, and movement intensity derived from accelerometer-related data.

We also included survey information such as end-of-study perceived stress and PHQ-9 features. Ecological momentary assessment variables were summarized into mean values for loneliness, connectedness, isolation, positive affect, and negative affect. Missing values were handled with median imputation inside the machine-learning pipelines. The final target variable was a binary loneliness label derived from the end-of-study UCLA Loneliness Scale using a median split.

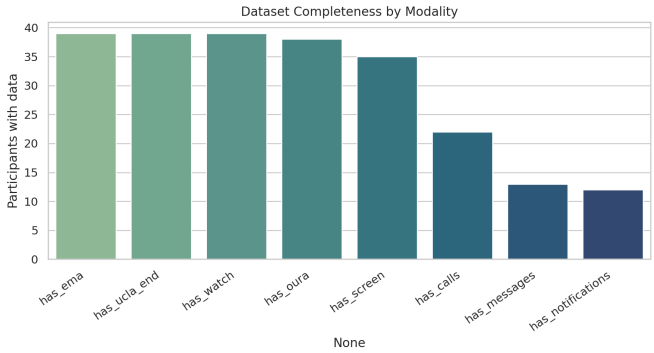

Fig. 1. Modality availability across participants.

III. Machine Learning Methods

We defined the prediction task as classifying participants into lower-versus-higher loneliness groups based on the final UCLA loneliness score. The label creation procedure was transparent and reproducible: UCLA survey items were scored, reverse-coded where appropriate, summed into a total loneliness score, and binarized using the sample median threshold.

Four supervised learning models were trained and compared according to the Project 4 description: SVM, Random Forest, XGBoost, and a Neural Network implemented with a multilayer perceptron. An average imputation was applied to all models, and scaling was used for the SVM and Neural Network. The data were split using an 80:20 stratified train/test

split, and 5-fold cross-validation with F1 scoring was used to reduce dependence on a single split.

IV. Results

Table I summarizes model performance on the held-out test set together with mean cross-validated F1 score.

TABLE I
MODEL PERFORMANCE COMPARISON

Model	Acc.	Prec.	Recall	F1	ROC-AUC	CV F1
Random Forest	0.8750	0.8000	1.0000	0.8889	0.7500	0.7569
XGBoost	0.7500	0.6667	1.0000	0.8000	0.7500	0.7159
SVM	0.5000	0.5000	1.0000	0.6667	0.1875	0.6993
Neural Network	0.5000	0.5000	1.0000	0.6667	0.4375	0.6993

Random Forest resulted in the best overall performance. It achieved the highest held-out F1 score and the highest accuracy while still remaining competitive in cross-validation. XGBoost had the second best performance, while SVM and the Neural Network underperformed on the test split.

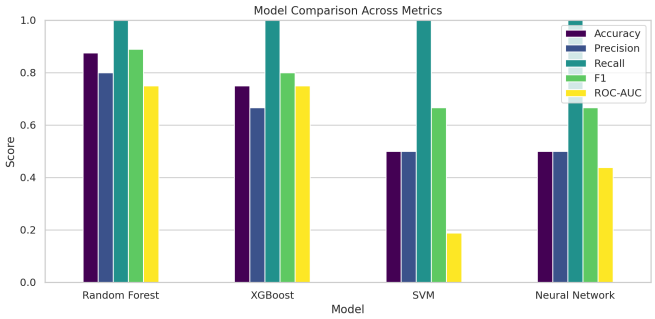

Fig. 2. Comparison of evaluation metrics across the four models.

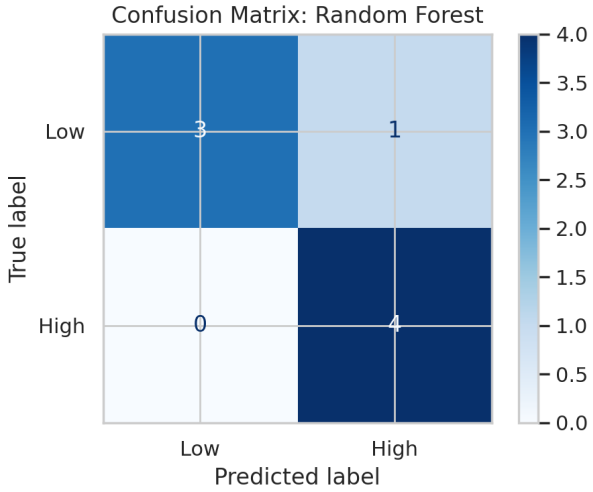

Fig. 3. Confusion matrix for the best-performing Random Forest model.

V. Discussion

Random Forest most likely performed best due to the dataset being relatively small and containing a heterogeneous mix of behavioral, physiological, and survey-derived variables. Additionally, tree-based ensemble models work well with nonlinear relationships and feature interactions, and are generally more stable than neural networks when sample size is limited. XGBoost was also competitive, which supports the claim that tree-based methods match this problem well.

Feature importance analysis showed that the strongest predictors were largely drawn from ecological momentary assessment and sleep-related features. For example, the Random Forest model's top predictors included positive affect, Oura sleep RMSSD, loneliness, negative affect, and connectedness, which suggests that both internal emotional state and sleep-related physiological recovery are closely tied to loneliness in this dataset.

The greater insight is that loneliness is not captured by a single modality, but rather, the prediction improves when subjective psychological signals are combined with passive behavioral and wearable-based measures. This supports the multimodal framing of the project and highlights the value of combining self-report, smartphone, and wearable data when studying mental health.

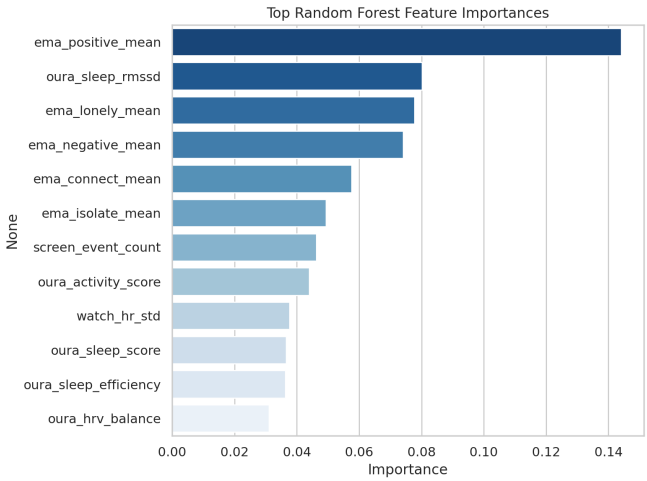

Fig. 4. Top Random Forest feature importances.

VI. Limitations

This project certainly has several limitations. For one, the sample size is very small, so test-set results may vary across different splits, something that proved true as some modalities were incomplete and required imputation. Additionally, the inclusion of EMA, PSS, and PHQ-derived variables improves predictive power but makes the task less purely passive than a sensor-only approach. Finally, the smartwatch heart-rate variability feature is only a proxy rather than a full beat-to-beat HRV reconstruction.

VII. Conclusion

This project shows that multimodal health data has the potential to predict loneliness-related outcomes with reasonable success. Out of the four models tested, Random Forest exhibited the best performance and identified both emotionally

and physiologically meaningful predictors. The results that we received support the idea that combining multiple behavioral and health-related data sources can improve machine-learning analysis of mental well-being.